

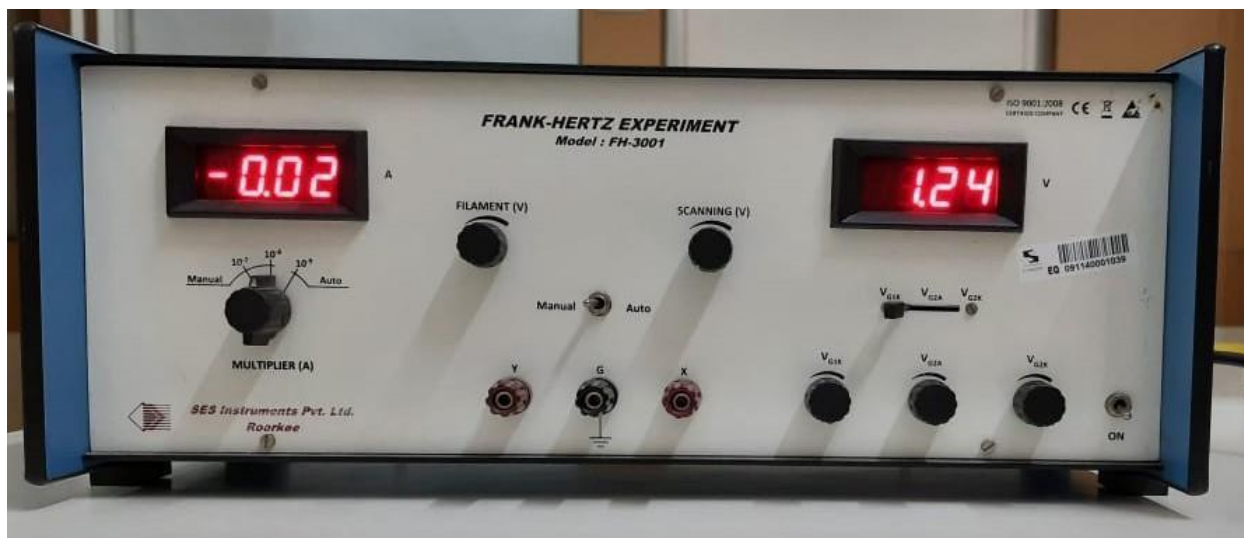
Experiment No. 10

FRANCK HERTZ EXPERIMENT

Aim: To experimentally demonstrate the concept of quantization of energy levels of atom (Bohr's model) using Franck-Hertz experiment and determine first excitation potential of Argon gas.

Apparatus: Tetrode tube filled with experimental Argon gas, filament, power supply three variable voltage sources, nanometer,

All above apparatus is in the following Franck Hertz Box.



Theory:

There was an experiment conducted by James Franck and Gustav Hertz in 1914, to demonstrate the existence of excited states in mercury atoms. It confirms the prediction of Bohr model of quantum theory that electrons occupy only discrete, quantized energy states. For this great invention they have been awarded Nobel Prize in Physics in the year 1925.

From the early spectroscopic work, it is clear that atoms emit radiation at discrete frequencies. From Bohr's model, the frequency of the radiation ν is related to the change of energy levels through $\Delta E = h\nu$. It is then to be expected that transfer of energy to atomic electrons by any mechanism should always be in discrete amounts. One such mechanism of energy transfer is through inelastic scattering of low-energy electrons.

Franck and Hertz in 1914 set out to verify these considerations.

- (a) It is possible to excite atoms by low energy electron bombardment.
- (b) The energy transferred from electrons to the atoms always had discrete values.
- (c) The values so obtained for the energy levels were in agreement with spectroscopic results.

In this experiment excitation of Ar atoms is studied by inelastic collision with electrons. The electrons emitted from the indirectly heated cathode of a tetrode filled with argon vapors are accelerated. The excited atoms emit visible light that can be viewed directly. The excitation energy of argon is determined from the distance between the equidistant maxima of the electron current in variable opposing electric field.

Experimental Set-up:

The experimental set up involves a tube containing low pressure experimental gas fitted with four electrodes: an electron-emitting cathode (K), a mesh grid (G_1) for minimizing space charge effects a mesh grid (G_2) for acceleration, and an anode (A). The anode was held at a slightly negative electrical potential relative to the grid G_2 (although positive compared to the cathode), so that electrons had to have at least a corresponding amount of kinetic energy to reach it after passing the grid and thereby making the dips in the plate current more prominent. Instruments were fitted to measure the current passing between the electrodes, and to adjust the potential difference (voltage) between the cathode (negative electrode) and the accelerating grids Figure 1.

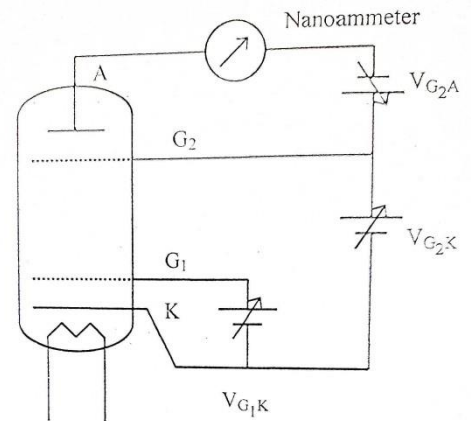


Figure 1: Circuit diagram of Franck-Hertz experiment

Operating Principle:

The Frank-hertz tube in this instrument is a tetrode filled with the argon vapour. Figure.1 shows the basic scheme of experiment.

The electrons emitted by filament can be accelerated by the potential V_{G2K} between the cathode

and the grid G_2 . The grid G_1 helps in minimizing space charge effects. The grids are wire mesh and allow the electrons to pass through. The plate A is maintained at a potential slightly negative with respect to the grid G_2 . This helps in making the dips in the plate current more prominent. In this experiment, the electron current is measured as a function of the voltage V_{G2K} . As Voltage increases, the electron energy goes up and so the electron can overcome the retarding potential V_{G2A} to reach the plate A. This gives rise to a current in the ammeter, which initially increases. As the voltage further increases, the electron energy reaches the threshold value to excite the atom in its first allowed excited state. In doing so, the electrons lose energy and therefore the number of electrons reaching the plate decreases. This decrease is proportional to the number of inelastic collisions that have occurred. When the V_{G2K} is increased further and reaches a value twice that of the first excitation potential, it is possible for an electron to excite an atom halfway between the grids, lose all its energy, and then gain a new enough energy to excite another atoms resulting in a second dip in the current. The advantage of this type of configuration of the potential is that the current dips are much more pronounced, and it is easy to obtain six-fold or even larger multiplicity in the excitation of the first level i.e. one can get 6 peaks (dips) or more.

Formula Used:

If V_n is the potential corresponding to n^{th} peak and V_1 is the potential corresponding to 1^{st} peak, then

$$\text{The } 1^{\text{st}} \text{ excitation potential} = \frac{V_n - V_1}{n-1} \dots (1)$$

Where $(n-1)$ is number of dips between 1^{st} and n^{th} peaks. Figure 2 shows plate current Vs V_{G2K} , for reference.

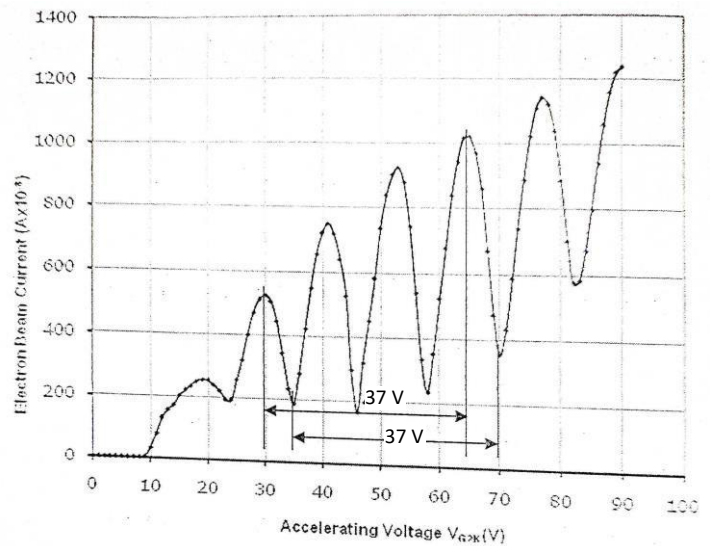


Figure 2

Experiment consists of the following:

- Argon filled tetrode
- Filament Power Supply: 2.6 – 3.3 V continuously variable
- Power Supply for V_{G1K} : 1.3 – 5 V continuously variable
- Power Supply for V_{G2A} : 1.3 – 15 V continuously variable
- Power Supply for V_{G2K} : 0 – 95 V continuously variable

Multirange Analogue Voltmeter Range: V_{G1K} : 0-5 V, V_{G2A} : 0-15 V & V_{G2K} : 0-100 V

Multirange Digital Ammeter Range: 0 – 100 V, 0 – 10 μ A & 0 – 1 μ A

Range Multiplier: 10^{-6} , 10^{-7} , 10^{-8} & 10^{-9}

The instrument can lead to a plot of the amplitude spectrum curve by means of point by point measurement.

Procedure

1. Before the power is switched 'ON' make sure all the control knobs are at their minimum position and Current Multiplier knob at 10^{-7} or 10^{-8} or 10^{-9} (whichever suitable) position.
2. Switch 'ON' the power.
3. Turn the manual- Auto Switch to manual and check that the Scanning Voltage Knob is at its minimum position.
4. Turn Voltage Display Selector to V_{G1K} and adjust the V_{G1K} knob until voltmeter reads 1.5V.
5. Turn Voltage display selector to V_{G2A} and adjust the V_{G2A} knob until the voltmeter reads 7.5V.

When you have finished step 1-5, you are ready to do the experiment.

Filament Voltage	: 2.6 V (minimum position)
V_{G1K}	: 1.5 V
V_{G2A}	: 7.5 V
V_{G2K}	: 1 – 95 V (Variable)
Current Multiplier	: 10^{-7} or 10^{-8}

6. Rotate V_{G2K} knob and observe the variation of plate current I_p with the increase of V_{G2K} voltage. The current reading would show maxima and minima periodically. The magnitude of maxima could be adjusted suitably by adjusting the filament voltage and the value of Current Multiplier. Now take the systematic readings, V_{G2K} Vs Plate current (I_p). For better resolution, the reading may be taken at an interval of 1 V (1/2 division).
7. Plot the graph with output current I_p on Y-axis and accelerating voltage V_{G2K} at X-axis.

Observation Table:

S. N.	Acceleration Potential V_{G2K} (Volts)	Plate Current I_p (mili Amperes)
1		
2		
3		
.		
.		

1. At low potential differences up to 12 volts when the tube contained argon vapour the current through the tube increased steadily with increasing potential difference. The higher voltage increased the electric field in the tube and electrons were drawn more forcefully towards and through the accelerating grid.
2. At 12 volts the current drops sharply, almost back to zero.
3. The current increases steadily once again if the voltage is increased further, until 24 volts is reached (exactly 12+12 volts).
4. At 24 volts a similar sharp drop is observed.

Calculation: Calculate the 1st excitation potential using equation (1)

Result:

1. Plot the Graph between output current on Y-axis and accelerating voltage V_{G2K} at X-axis. This shows series of dips in current at approximately 12 volt (say) increments. A graph similar to given in Figure 2.
2. The First Excitation potential is

Precautions

1. During the experiment (manual), when the voltage V_{G2K} is over 60 V, please pay attention to the output current indicator, if the ammeter reading increase suddenly, decrease the voltage at once to avoid the damage of the tube.

2. If you want to change the value of V_{G1K} , V_{G2A} and Filament Voltage during experiment, please first adjust the value of V_{G2K} to 'Zero'.
3. Whenever the filament voltage is changed, please allow 2-3 minutes for its stabilization.
4. When the Frank-Hertz Tube is already in the socket, please make sure the following before the power is switched 'ON' or 'OFF', to avoid damage to the tube.
5. Manual - Auto switch is on Manual and Scanning and Filament Voltage knob at its minimum position (rotate it anticlockwise) and current multiplier knob at 10^{-7} .